

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

Finding Reasoning Paths :

- 1 . **Delayed or absent menarche** â‡‡ hypothalamic-pituitary-gonadal axis dysfunction â‡‡ pituitary dysfunction â‡‡ increased prolactin levels â‡‡ galactorrhea .
- 2 . **Bilateral spontaneous milky nipple discharge** â‡‡ galactorrhea â‡‡ prolactinoma or other pituitary pathology â‡‡ decreased gonadotropin release (FSH and LH).
- 3 . **Delayed or absent menarche** â‡‡ hypogonadotropic hypogonadism â‡‡ pituitary dysfunction â‡‡ elevated prolactin levels â‡‡ galactorrhea .
- 4 . **Bilateral spontaneous milky nipple discharge** â‡‡ increased prolactin levels â‡‡ galactorrhea â‡‡ gonadotropin suppression (FSH , LH) â‡‡ hypogonadotropic hypogonadism .

Reasoning Process :

1 . **Patient Presentation** :

- The patient is a 34-year-old woman presenting with delayed or absent menarche , missed periods , and bilateral spontaneous milky nipple discharge . These symptoms suggest an underlying hormonal imbalance , likely involving the pituitary gland .
- The vital signs are stable , and the neurological examination is normal , which reduces the likelihood of a primary neurological cause for the symptoms .

2 . **Differential Diagnosis** :

- **Prolactinoma** : A prolactin-secreting pituitary adenoma can cause hyperprolactinemia , leading to galactorrhea and secondary hypogonadism (reduced FSH and LH release).
- **Spontaneous Pituitary Hyperplasia** : This is another possibility for increased prolactin secretion , though it is less common than a prolactinoma .
- **Hypothalamic Pituitary Disease** : Conditions such as shear traction or other lesions in the hypothalamus or pituitary gland can also cause elevated prolactin levels and hypogonadotropic hypogonadism .

3 . **Pathophysiology** :

- In cases of pituitary hyperplasia or prolactin-secreting adenomas , increased prolactin levels suppress gonadotropin-releasing hormone (GnRH) secretion .
- This suppression leads to decreased release of FSH and LH from the anterior pituitary , resulting in hypogonadotropic hypogonadism .
- The hypogonadotropic hypogonadism causes secondary amenorrhea and galactorrhea .

4 . **Expected Laboratory Findings** :

- **Increased Prolactin Levels** : Due to a prolactinoma or spontaneous pituitary hyperplasia .
- **Decreased FSH and LH** : Due to gonadotropin suppression by increased prolactin levels .
- **Normal Estradiol Levels** : In the absence of primary ovarian failure , estradiol levels are maintained .
- **No Evidence of Central Hypothyroidism** : The absence of thyroid-stimulating hormone (TSH) suppression indicates that thyroid function is normal .
- **No Premature Ovarian Failure** : The delay in menarche and subsequent amenorrhea suggest hypogonadotropic hypogonadism rather than premature ovarian failure .

5 . **Exclusion of Other Causes** :

- Hypothalamic mechanisms (e.g. , Sheehan syndrome , head injury , or pituitary stalk interruption) are less likely here because the pituitary function is normal , and there is no evidence of TSH suppression .
- MNGIT mutations are unlikely in this case because the symptoms and physical findings are more consistent with pituitary pathology .

Conclusion :

The patient's laboratory findings would be expected to show **increased prolactin levels , decreased FSH , decreased LH , normal estradiol , and no evidence of central hypothyroidism or ovarian failure** . This